Agricultural Irrigation Data Analysis

Name -Divash Kumar Yadav
Registration No: 12406109
Section -K2M030
Programme B.Tech. in CSE
Lovely Professional University
Phagwara ,Punjab 144411

School of Computer Science and Engineering

Abstract - This study presents a comprehensive analysis of agricultural irrigation data using Python-based data science techniques. The dataset contains detailed information on irrigated areas across different regions, sources of irrigation such as canals, wells, and tanks, and yearly agricultural trends.

The analysis is conducted using Python libraries including Pandas for data manipulation, NumPy for numerical operations, and Matplotlib and Seaborn for visualization. The methodology involves data preprocessing, exploratory data analysis (EDA), statistical analysis, and predictive modeling using machine learning techniques.

Key analyses include the distribution of irrigated land, comparison of irrigation sources, trend analysis across years, and correlation between net and gross irrigated areas. The study also applies linear regression to predict irrigation patterns.

The results reveal significant relationships between irrigation sources and agricultural productivity, highlighting the importance of efficient water management. This study demonstrates how data-driven approaches can support better decision-making in agriculture and irrigation planning.

I. INTRODUCTION

Agriculture is one of the most important sectors contributing to economic growth and food security. Irrigation plays a crucial role in enhancing crop productivity by ensuring a reliable water supply.

Agricultural irrigation involves the artificial application of water to land for assisting crop growth. Different sources of irrigation such as canals, tubewells, tanks, and other systems are used depending on geographical and climatic conditions.

This study focuses on analyzing irrigation data across different regions and years. The dataset includes multiple attributes such as net irrigated area, gross irrigated area, and irrigation sources.

With the help of data science techniques, large irrigation datasets can be effectively analyzed to identify trends and patterns. This project uses Python tools such as Pandas, NumPy, Matplotlib, and Seaborn to perform data cleaning, visualization, and analysis.

The insights obtained from this analysis help in understanding irrigation efficiency, regional differences, and the impact of water resources on agricultural productivity.

II. SOURCE OF DATASET

Country	State	District	Year	Net_Govt	Net_Pvt_C	Net_Total	Net_Tank	Net_Tube	Net_Other	Net_Total	Net_Other	Net_Total	Gross_Gov	Gross_Pvt	Gross_Tot	Gross_Tan	Gross_Tub	Gross_Oth	Gross_Tot	Gross_Oth Gr
India	Andaman	Nicobars	2022																	
India	Andaman	North And	2022																	
India	Andaman	South And	2022																	
India	Andhra Pr	Alluri Sitha	2022	3733		3733	1838	522		522	19824	25917	6030		6030	2035	842		842	22158
India	Andhra Pr	Anakapalli	2022	42638		42638	18608	15184	203	15387	666	77299	43871		43871	19288	19808	240	20048	684
India	Andhra Pr	Ananthapu	2022	25963		25963	1230	157892		157892	245	185330	26663		26663	1330	183310		183310	245
India	Andhra Pr	Annamayy	2022	53		53	749	65947	1888	67835		68637	55		55	781	83615	2047	85662	
India	Andhra Pr	Bapatla	2022	124992		124992	167	21296	61	21357	2984	149500	148886		148886	167	35436	61	35497	3165
India	Andhra Pr	Chittoor	2022	22		22	1863	72438	1867	74305		76190	22		22	2033	86403	2109	88512	
India	Andhra Pr	Dr. B.R. Ar	2022	71003		71003	260	19078		19078	441	90782	124770		124770	593	34097		34097	880
India	Andhra Pr	East Godav	2022	22157		22157	13398	71258		71258	13600	120413	42636		42636	15532	118410		118410	20221
India	Andhra Pr	Eluru	2022	37412		37412	14310	123416	133	123549	2393	177664	53991		53991	14425	180731	171	180902	2996
India	Andhra Pr	Guntur	2022	80208		80208	307	9571	203	9574	3037	93126	97505		97505	307	27266	250	27516	3136
India	Andhra Pr	Kakinada	2022	71195		71195	9218	17923	6	17929	11359	109701	131233		131233	10639	24056	22	24078	11955
India	Andhra Pr	Krishna	2022	149557		149557	3293	19005	55	19060	1331	173241	154012		154012	3479	33963	130	34093	1379
India	Andhra Pr	Kurnool	2022	28979		28979	990	61074	496	61570	8249	99788	35230		35230	1130	80384	496	80880	11756
India	Andhra Pr	Nandyal	2022	81771		81771	9368	70893	8988	79881	9819	180839	106993		106993	10524	92799	10644	103443	11257
India	Andhra Pr	Ntr	2022	23497		23497	14801	22545	1134	23279	9174	70751	32445		32445	16276	35707	1995	37702	12656
India	Andhra Pr	Palnadu	2022	100875		100875	626	19052	1110	20162	3855	125518	114714		114714	626	19262	1119	20381	3935
India	Andhra Pr	Parvathipu	2022	35500		35500	19657	22478	126	22604	3926	81687	36097		36097	19976	27448	180	27628	4207
India	Andhra Pr	Prakasam	2022	16328		16328	2948	77250	58	77308	1379	97963	16389		16389	2948	77338	58	77396	1379
India	Andhra Pr	Sri Potti Sr	2022	109383		109383	13440	48239	565	48804	200	171827	130697		130697	13658	53684	565	54249	205
India	Andhra Pr	Sri Sathya	2022	5		5	1152	87030	152	87182		88339	5		5	1258	103327	152	103479	
India	Andhra Pr	Srikakulam	2022	88803		88803	42158	2346	779	3125	2598	136684	101054		101054	46166	11369	11250	22619	2749
India	Andhra Pr	Tirupati	2022	7499		7499	47240	59456	3163	62619	1930	119288	7575		7575	48728	70841	3704	74545	1930
India	Andhra Pr	Visakhapat	2022	792		792	2451	2275	383	2658	6	5907	792		792	2725	3860	415	4275	6
India	Andhra Pr	Vizianagar	2022	35636		35636	53041	15024	2927	17951	1411	108039	39628		39628	59377	26431	5144	31575	1850
India	Andhra Pr	West Godi	2022	86171		86171	1343	10138		10138	567	98438	166853		166853	2684	18863		18863	1092

1. The dataset used in this project is sourced from the National Data & Analytics Platform (NDAP), developed by NITI Aayog, Government of India.
2. The specific dataset analyzed is related to **Agricultural Irrigation Statistics**, available at: https://ndap.niti.gov.in/dataset/6796
3. This dataset provides detailed information on irrigation patterns across different states and districts in India, covering multiple years.
4. The dataset includes key attributes such as:
    - Net irrigated area
    - Gross irrigated area
    - Irrigation sources (canals, tubewells, tanks, wells, etc.)
    - State and district-level data
    - Year-wise agricultural information
5. The data is provided in a structured tabular format (CSV), making it suitable for analysis using tools like Python, Excel, or R.
6. The dataset is publicly available under the Open Government Data (OGD) initiative, enabling researchers and students to perform data-driven analysis for academic and research purposes.
7. The dataset is reliable and regularly updated, ensuring accuracy and relevance for agricultural and irrigation analysis.

III. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is a crucial step in understanding the dataset before performing advanced analysis or modeling. In this project, EDA was conducted to examine the structure, quality, and patterns within the agricultural irrigation dataset.

A. Data Inspection

The dataset was loaded into a Pandas DataFrame using the `read_csv()` function. Initial inspection was performed using:

- `head()` to view sample records
- `info()` to understand data types
- `shape` to determine rows and columns

This helped in gaining a basic understanding of the dataset.

B. Handling Missing Values

Missing values were identified using:

- `isnull().sum()`

Observations:

- Most important columns had minimal or no missing values
- Few non-critical columns contained null values

Action Taken:

- Forward fill method (`ffill()`) was used where required
- Non-essential missing data was ignored for analysis

C. Data Cleaning

Data cleaning included:

- Removing unnecessary columns
- Renaming columns for clarity
- Converting data types where required

This ensured consistency and improved data usability.

D. Outlier Detection

Outliers were analyzed using:

- Box plots
- Summary statistics

Findings:

- Some extreme values were observed in irrigation data
- These represent highly irrigated regions and were retained

E. Summary Statistics

Basic statistical analysis was performed using:

- `describe()` function

This provided:

- Mean
- Median
- Standard deviation
- Minimum and maximum values

These statistics helped in understanding data distribution.

F. Key Insights from EDA

- Dataset is largely clean and structured
- Irrigation data shows variation across regions
- Some states dominate in irrigation usage
- Strong relationships exist between irrigation variables

IV. Dataset Structure and Summary Statistics

After ensuring that the dataset was clean and complete, the next step was to understand its structure and statistical properties.

A. Dataset Overview with info():

The `info()` function was used to:

1. Display the total number of entries and columns.
2. Identify data types (e.g., int64, object) for each column.
3. Check memory usage and null values.

```
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Country                 17626 non-null  object
 1   State                   17626 non-null  object
 2   District                17626 non-null  object
 3   Year                    17626 non-null  int64
 4   Net_Govt_Canal          17626 non-null  float64
 5   Net_Pvt_Canal           17626 non-null  float64
 6   Net_Total_Canal         17626 non-null  float64
 7   Net_Tank                17626 non-null  float64
 8   Net_Tubewell            17626 non-null  float64
 9   Net_Other_Well          17626 non-null  float64
 10  Net_Total_Well          17626 non-null  float64
 11  Net_Other_Source        17626 non-null  float64
 12  Net_Total_Irrigated     17626 non-null  float64
 13  Gross_Govt_Canal        17626 non-null  float64
 14  Gross_Pvt_Canal         17626 non-null  float64
 15  Gross_Total_Canal       17626 non-null  float64
 16  Gross_Tank              17626 non-null  float64
 17  Gross_Tubewell          17626 non-null  float64
 18  Gross_Other_Well        17626 non-null  float64
 19  Gross_Total_Well        17626 non-null  float64
 20  Gross_Other_Source      17626 non-null  float64
 21  Gross_Total_Irrigated   17626 non-null  float64
```

B. Findings:

URL: https://ndap.niti.gov.in/dataset/6796

1. The dataset consists of both numerical and categorical columns.
2. Important attributes like Area, Year, and Value are numeric and suitable for quantitative analysis.
3. Categorical variables such as Area Name and Item describe irrigation-related categories.

C. Statistical Summary with describe():

The `describe()` function provided the following for numerical columns:

1. Count – Total number of non-null values
2. Mean – Average value
3. Standard Deviation (std) – Measure of variation
4. Min/Max – Range of values
5. 25%, 50%, 75% Quartiles – Distribution of the data

```
In [185]: print(df.describe())
                Year  Net_Govt_Canal  ...  Gross_Other_Source  Gross_Total_Irrigated
count  17626.000000    17626.000000  ...        17626.000000           17626.000000
mean    2010.227164    25983.343867  ...         7170.828323               0.121002
std        7.216838    45427.125429  ...        22672.540264               0.113878
min     1998.000000        1.000000  ...            1.000000               0.000000
25%     2004.000000     8584.500000  ...         3029.000000               0.049656
50%     2010.000000    12774.000000  ...         3041.500000               0.096997
75%     2017.000000    18915.000000  ...         3052.500000               0.148104
max     2022.000000   652031.000000  ...       606800.000000               1.000000
```

D. Findings:

1. Significant variation was observed in irrigation values across different years and areas.
2. Some variables showed high maximum values compared to the median, indicating skewed distribution.
3. The dataset provides meaningful insights into irrigation trends and variability.

E. Conclusion:

Using `info()` and `describe()`, the dataset was confirmed to be:

1. Structurally well-organized and suitable for analysis.
2. Containing variability that reflects real-world differences in agricultural irrigation patterns.

Objective 1: Top 5 States with Maximum Average Net Irrigated Area

i. Introduction

This objective aims to identify the top 5 states with the highest average Net Irrigated Area and analyze their trends over different years.

ii. General Description

The dataset is grouped by state to calculate the average Net Total Irrigated Area. The top 5 states are selected based on highest average values. A line plot is used to visualize how irrigation trends change over time for these states.

iii. Specific Requirements, Functions, and Formulas

- **Grouping:**
  `groupby('State')` – groups data by state

- **Aggregation:**
  `mean()` – calculates average Net Irrigated Area

- **Top Selection:**
  `nlargest(5)` – selects top 5 states

- **Filtering:**
  `isin()` – filters dataset for selected states

- **Visualization:**
  `plt.plot()` – line graph for yearly trend

Code

```
# Line Plot - Top 5 States average Net Irrigated Area over Years
top5 = df.groupby('State')['Net_Total_Irrigated'].mean().nlargest(5).index
df_top5 = df[df['State'].isin(top5)]


plt.figure(figsize=(10, 5))
for state in top5:
    state_data = df_top5[df_top5['State'] == state].groupby('Year')['Net_Total_Irrigated'].mean()
    plt.plot(state_data.index, state_data.values, marker='o', label=state)

plt.title('Average Net Irrigated Area Over Years (Top 5 States)')
plt.xlabel('Year')
plt.ylabel('Net Total Irrigated (Normalized)')
plt.legend()
plt.grid()
plt.xticks(range(1998, 2023, 2), rotation=45)
plt.tight_layout()
plt.show()
```

iv. Analysis Results

- Top 5 states with highest irrigation were identified

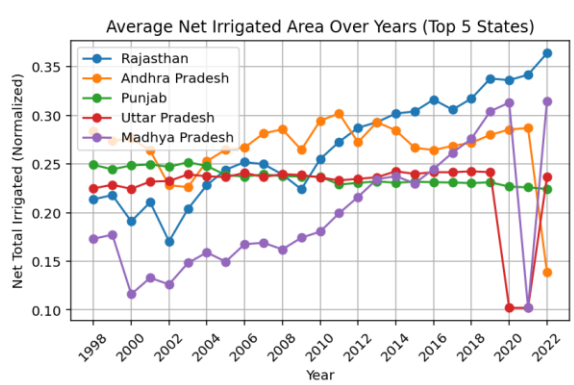
- These states show consistent irrigation patterns over the years

- Some states show increasing trends while others remain stable

- The visualization clearly highlights differences in irrigation levels among states

v. Visualization

A line plot is used to show the trend of average Net Irrigated Area over years for the top 5 states.
Each line represents a state, making it easy to compare irrigation growth and patterns.



Objective 2: Share of Irrigation Sources using Pie Chart

i. Introduction

This objective aims to analyze the contribution of different irrigation sources to the total irrigated area.

ii. General Description

The total irrigation area from various sources such as canals, wells, tanks, and other sources is calculated. A pie chart is used to represent the percentage share of each irrigation source.

iii. Specific Requirements, Functions, and Formulas

• **Aggregation:**
`sum()` – calculates total irrigation from each source

• **Column Selection:**
`[['Net_Total_Canal', 'Net_Total_Well', 'Net_Tank', 'Net_Other_Source']]`

• **Visualization:**
`plot.pie()` – creates pie chart

• **Percentage Display:**
`autopct='%1.1f%%'` – shows percentage values

Code .

```
# Pie Chart - Share of Irrigation Sources
source_totals = df[['Net_Total_Canal', 'Net_Total_Well', 'Net_Tank', 'Net_Other_Source']].sum()
source_totals.plot.pie(autopct='%1.1f%%', figsize=(6, 6))
plt.title('Share of Irrigation Sources')
plt.ylabel('')
plt.show()
```

iv. Analysis Results

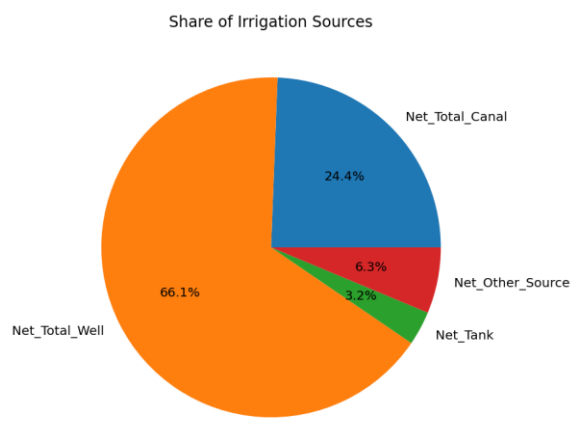
• Wells and canals contribute the largest share of irrigation

• Tanks and other sources contribute less

• Clear distribution of irrigation dependency is observed

v. Visualization

A pie chart represents the percentage contribution of each irrigation source, making comparison simple and visually clear.



Objective 3: Relationship between Net and Gross Irrigated Area using Scatter Plot

i. Introduction

This objective aims to analyze the relationship between Net Total Irrigated Area and Gross Total Irrigated Area.

ii. General Description

A scatter plot is used to visualize how Net Irrigated Area and Gross Irrigated Area are related. Each point represents a data record, helping to understand the pattern and dependency between the two variables.

iii. Specific Requirements, Functions, and Formulas

- **Visualization Tool:**
  `seaborn.scatterplot()` – used to create scatter plot

- **X-axis:**
  `Net_Total_Irrigated`

- **Y-axis:**
  `Gross_Total_Irrigated`

- **Plot Settings:**
  `alpha=0.4` – controls transparency
  `color='green'` – sets color

Code.

```
# Scatter Plot - Net vs Gross Irrigated Area
plt.figure(figsize=(8, 4))
sns.scatterplot(x='Net_Total_Irrigated', y='Gross_Total_Irrigated', data=df, alpha=0.4, color='gre
plt.title('Net vs Gross Irrigated Area')
plt.xlabel('Net Total Irrigated')
plt.ylabel('Gross Total Irrigated')
plt.show()
```

iv. Analysis Results

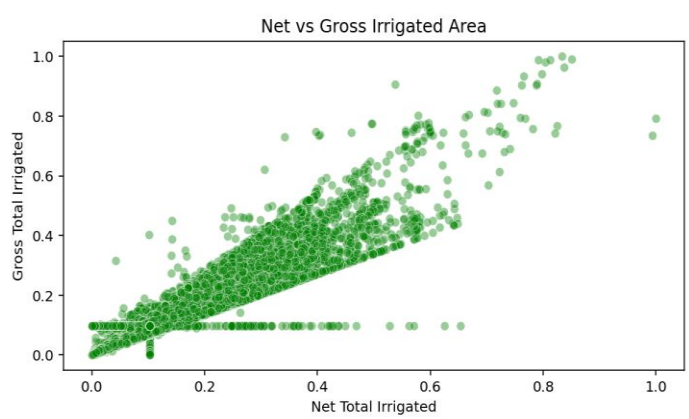
- A strong positive relationship is observed between Net and Gross Irrigated Area

- As Net Irrigated Area increases, Gross Irrigated Area also increases

- Data points follow an upward trend, indicating dependency between variables

v. Visualization

The scatter plot clearly shows the correlation between Net and Gross Irrigation.
It helps in identifying patterns and understanding how one variable influences the other.



Objective 4:

Correlation Analysis of Irrigated Area using Heatmap

i. Introduction

This objective aims to analyze the correlation between different irrigation variables to understand their relationships.

ii. General Description

A correlation matrix is created using selected irrigation features such as Net Total Irrigated Area, Gross Total Irrigated Area, Tubewell, and Tank irrigation. A heatmap is used to visually represent the strength of relationships between these variables.

iii. Specific Requirements, Functions, and Formulas

• **Correlation Calculation:**
`corr()` – computes correlation between variables

• **Selected Features:**
`Net_Total_Irrigated, Gross_Total_Irrigated, Net_Tubewell, Net_Tank, Gross_Tubewell`

• **Visualization:**
`sns.heatmap()` – displays correlation matrix

• **Parameters:**
`annot=True` – shows correlation values
`cmap='coolwarm'` – color scale
`linewidths=0.5` – grid lines

iv. Analysis Results

• Strong positive correlation observed between Net and Gross Irrigated Area

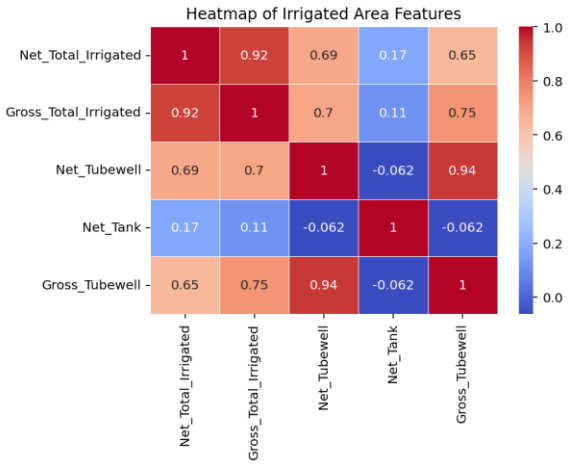
• Tubewell irrigation shows moderate relationship with total irrigation

• Tank irrigation shows weaker correlation compared to other variables

• Heatmap clearly highlights relationships between all selected features

v. Visualization

The heatmap provides a clear visual representation of correlations.
Darker colors indicate stronger relationships, making it easy to identify key influencing variables.



Objective 5: Exploratory Data Analysis and Statistical Analysis of Irrigation Data

---

i. Introduction

This objective aims to analyze the statistical properties and distribution of irrigation data to understand patterns, relationships, and data behavior.

ii. General Description

Various statistical techniques are applied to examine irrigation variables such as Net and Gross Irrigated Area. The analysis includes summary statistics, distribution measures, correlation, outlier detection, and hypothesis testing.

iii. Specific Requirements, Functions, and Formulas

• **Summary Statistics:**
`describe()` – provides mean, std, min, max values

• **Skewness & Kurtosis:**
`skew(), kurt()` – measure distribution shape

• **Correlation:**
`corr()` – identifies relationships between variables

• **Outlier Detection:**
IQR method → `(Q1 - 1.5*IQR, Q3 + 1.5*IQR)`

• **Hypothesis Testing:**
`t-test (stats.ttest_ind)` – compares canal vs tubewell irrigation

• **Normality Test:**
`Shapiro-Wilk test` – checks data distribution

• **Visualization:**
`sns.pairplot()` – shows relationships between variables

CODE.

```
# Skewness and Kurtosis
print("Skewness:\n", df[numeric_cols].skew())
print("Kurtosis:\n", df[numeric_cols].kurt())

# Correlation
print("Correlation:\n", df[['Net_Total_Irrigated', 'Gross_Total_Irrigated', 'Net_Tubewell']].corr())

# Outlier Detection using IQR
Q1 = df['Net_Total_Irrigated'].quantile(0.25)
Q3 = df['Net_Total_Irrigated'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
outliers = df[(df['Net_Total_Irrigated'] < lower_bound) | (df['Net_Total_Irrigated'] > upper_bound)]
print("Outliers found:", len(outliers))
```

iv. Analysis Results

• Data shows variation and slight skewness in irrigation values

• Strong correlation observed between Net and Gross Irrigated Area

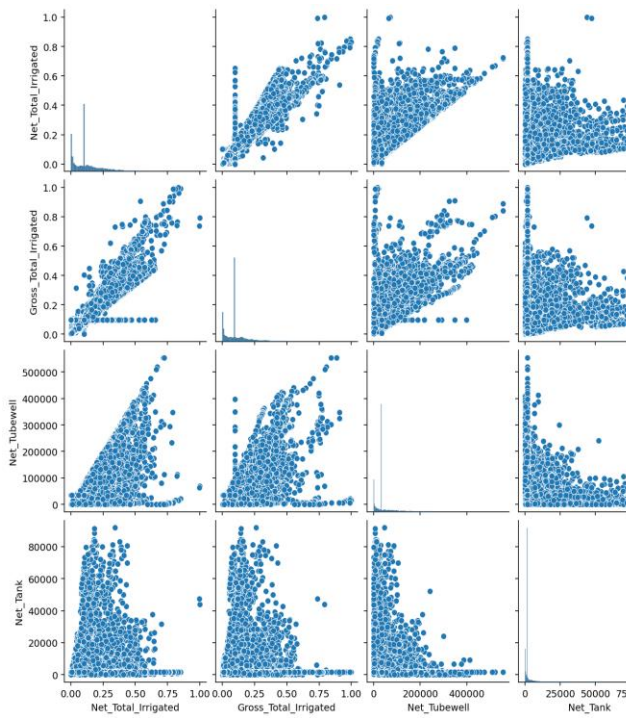
• Outliers are present, indicating extreme irrigation values in some regions

• T-test shows whether canal and tubewell irrigation differ significantly

• Shapiro test indicates whether data follows normal distribution

v. Visualization

Pair plot provides a visual understanding of relationships between irrigation variables.
It helps in identifying patterns, clusters, and correlations in the dataset.



Objective 6: Prediction of Gross Irrigated Area using Linear Regression

---

i. Introduction

This objective aims to predict Gross Total Irrigated Area using Net Total Irrigated Area with the help of a linear regression model.

ii. General Description

The dataset is divided into training and testing sets. A Linear Regression model is trained using Net Irrigated Area as input and Gross Irrigated Area as output.

iii. Specific Requirements, Functions, and Formulas

• **Feature Selection:**

```
x = Net_Total_Irrigated
y = Gross_Total_Irrigated
```

• **Train-Test Split:**

`train_test_split()` – splits data into training and testing sets

• **Model:**

`LinearRegression()` – used for prediction

• **Training:**

`model.fit()` – trains the model

• **Prediction:**

`model.predict()` – predicts values

Code.

```
# Predict Gross Total Irrigated from Net Total Irrigated
x = df[['Net_Total_Irrigated']]
y = df['Gross_Total_Irrigated']

# Train-Test Split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_st

# Model
model = LinearRegression()
model.fit(x_train, y_train)

# Predict
y_pred = model.predict(x_test)

# Scatter + Regression Line
plt.figure(figsize=(8, 4))
plt.scatter(x, y, color='blue', alpha=0.3)
plt.plot(x, model.predict(x), color='red', linewidth=2)
plt.xlabel('Net Total Irrigated (Normalized)')
plt.ylabel('Gross Total Irrigated (Normalized)')
plt.title('Linear Regression: Net vs Gross Irrigated Area')
plt.show()
```

iv. Analysis Results

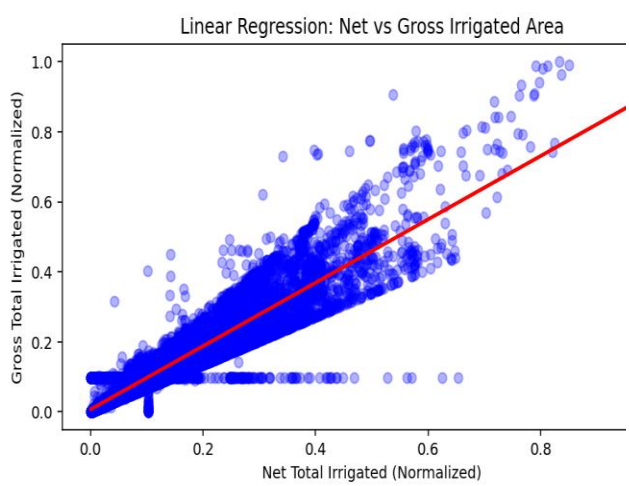
• Model successfully predicts Gross Irrigated Area

• Shows strong relationship between Net and Gross irrigation

• Prediction results indicate a linear trend

v. Visualization

A regression line is plotted over scatter data to show the relationship between Net and Gross Irrigated Area.



Objective 7: Evaluation of Regression Model Performance

i. Introduction

This objective evaluates the performance of the regression model using standard evaluation metrics.

ii. General Description

The model predictions are compared with actual values to measure accuracy using error metrics.

iii. Specific Requirements, Functions, and Formulas

• **Mean Squared Error (MSE):**
`mean_squared_error()`

• **R^2 Score:**
`r2_score()`

• **Mean Absolute Error (MAE):**
`mean_absolute_error()`

• **Custom Prediction:**
Model is used to predict value for input = 0.5

iv. Analysis Results

• Low MSE indicates good prediction accuracy

• R^2 score shows strong model performance

• MAE confirms minimal prediction error

• Model can predict new irrigation values effectively

v. Visualization

Evaluation is supported by regression graph and predicted outputs.

```
# Evaluation
mse = mean_squared_error(y_test, y_pred)
r2  = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
print(f"MSE : {mse:.4f}")
print(f"R2  : {r2:.4f}")
print(f"MAE : {mae:.4f}")
```

Result->

```
In [196]: print(f"MSE : {mse:.4f}")
MSE : 0.0020

In [197]: print(f"R2  : {r2:.4f}")
R2  : 0.8533

In [198]: print(f"MAE : {mae:.4f}")
MAE : 0.0258
```

Final Prediction Output

```
# Predict for a custom value
check = pd.DataFrame({'Net_Total_Irrigated': [0.5]})
result = model.predict(check)
print("Predicted Gross Irrigated for Net=0.5:", result)
```

Result->

```
In [203]: print("Predicted Gross Irrigated for Net=0.5:", resul
Predicted Gross Irrigated for Net=0.5: [0.45929799]
```

8.Conclusion

This project focused on analyzing agricultural irrigation data using data science techniques. The study involved data preprocessing, exploratory data analysis (EDA), visualization, statistical analysis, and machine learning.

The analysis revealed that irrigation patterns vary significantly across different states and over time. The line plot of top states showed clear trends in irrigation growth, indicating regional differences in agricultural development. The pie chart analysis highlighted that **well/tubewell irrigation contributes the largest share**, followed by canals, while tanks and other sources have relatively smaller contributions.

The scatter plot and heatmap analysis demonstrated a **strong positive relationship between Net and Gross Irrigated Area**, confirming that as net irrigation increases, gross irrigation also increases proportionally. The histogram and boxplot provided insights into data distribution and variability, showing the presence of skewness and outliers in certain states.

Further statistical analysis confirmed variations in irrigation data and identified significant relationships between irrigation features. The machine learning model (Linear Regression) achieved good performance with a high R^2 score, indicating that **Net Irrigated Area is a strong predictor of Gross Irrigated Area**. The prediction results further validated the linear relationship between these variables.

Overall, the project demonstrates how data science techniques can be effectively used to extract meaningful insights from agricultural data. These insights can support better decision-making in irrigation planning, resource allocation, and sustainable agricultural development.

9.Future Scope

This project provides a basic analysis of agricultural irrigation data using data science and machine learning techniques. However, there are several areas where the work can be further improved and expanded.

In the future, more advanced machine learning models such as Decision Trees, Random Forest, and Neural Networks can be applied to improve prediction accuracy. Time-series forecasting techniques can also be used to predict irrigation trends for upcoming years.

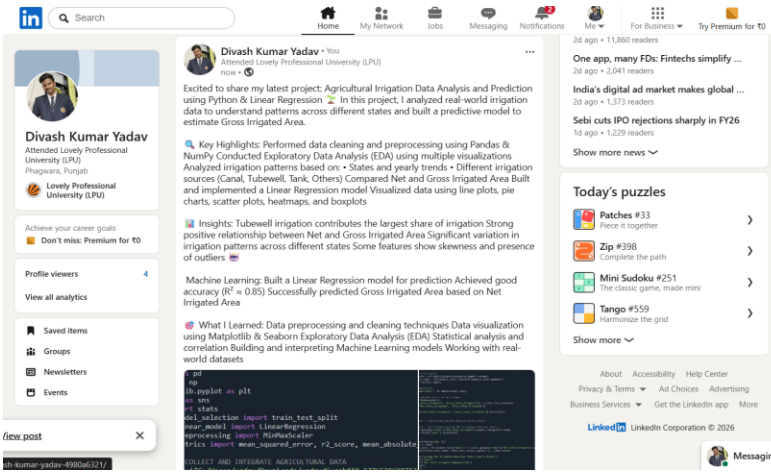
The dataset can be extended by integrating additional factors such as rainfall, soil type, temperature, and crop yield data. This would help in building more comprehensive and realistic models for agricultural analysis.

Real-time data collection using APIs and IoT-based sensors can also be incorporated to monitor irrigation systems dynamically. This would enable smart irrigation systems and better water resource management.

Additionally, the analysis can be enhanced by developing interactive dashboards using tools like Power BI or Tableau, making it easier for policymakers and farmers to understand the insights.

Overall, the future scope of this project lies in applying advanced analytics, integrating multiple data sources, and building intelligent systems for sustainable and efficient irrigation management.

10. References

1. NITI Aayog. *Agricultural Irrigation Dataset.* National Data & Analytics Platform (NDAP), Government of India. https://ndap.niti.gov.in/dataset/6796
2. NITI Aayog. *National Data & Analytics Platform (NDAP).* Government of India.
3. McKinney, W. (2010). *Data Structures for Statistical Computing in Python.* Proceedings of the 9th Python in Science Conference.
4. Hunter, J. D. (2007). *Matplotlib: A 2D Graphics Environment.* Computing in Science & Engineering.
5. Waskom, M. L. (2021). *Seaborn: Statistical Data Visualization.* Journal of Open Source Software.
6. Van Rossum, G., & Drake, F. L. (2009). *Python 3 Reference Manual.* Python Software Foundation.
7. Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python.* Journal of Machine Learning Research.

11.LinkedIn Post